

Why Our Feature Set Is Sufficient — and Why “Correlates With the Line” Is Not an Edge

Market efficiency, our two-model design, and the Infinity (Ivan_BP) variables · MLB Scion · 25 Jun 2026

1. The betting market is highly efficient — the central fact

For the games we bet (lines within ± 150), the bookie line already absorbs almost everything knowable about a matchup. The consequence is hard: the **predictable** signal left for **any** model is intrinsically small. Across the ~1,180 candidate inputs we tested, the most informative single one — the line itself — accounts for **under 1% of the uncertainty** in who wins; our best baseball statistic, about a third of one percent; every input's effect size is, by the standard statistical yardstick, “negligible.” It is like predicting a coin weighted by one or two percent: the bias is real and bettable, but tiny — and no cleverness in choosing what to measure can create more bias than physically exists.

Two conclusions follow, both reassuring: **our features are not the limitation** — we already use the strongest signals that exist, so adding more cannot help when there is no strong signal to find; and this is exactly why model accuracy sits near **52–53%** — close to the natural ceiling, not a shortfall. Profit comes from being right slightly more often than the price implies in the right situations and sizing bets well — not from a high accuracy the market makes impossible.

2. Our two strongest features

- **The bookie line** (home implied probability) — the market's own price, and our single most informative input (association with the result ~ 0.10 on a $-1 \dots +1$ scale). It is strong because it aggregates the collective information of the entire market.
- **MenOnBase_Strength (season-to-date)** — our best engineered statistic: how reliably a team gets runners on base. Of all ~1,170 non-line inputs it ranks **#1** by association with the actual result (~ 0.058 — modest in absolute terms, the hallmark of an efficient market, but the best clean signal we have). It does move with the line (~ 0.50 on a $0-1$ scale) — unsurprising, since good offences get favourable prices — but **we built it ourselves from box-score data, so we know exactly what it is, that it carries no line information, and that it cannot leak**. That auditability is what makes it trustworthy.

3. Why we run two models, not one

LineStats (V119) uses the line plus statistics — anchored to the market consensus. **StatsOnly (V123)** is deliberately blind to the line — built from baseball statistics only. When a market-anchored model and a market-blind model independently agree, that agreement is far more trustworthy than either alone — like two independent diagnostic methods reaching the same conclusion. We keep StatsOnly genuinely line-free precisely so its agreement is real confirmation, not an echo of the price.

4. Why the Infinity / Ivan_BP variables are high-risk — not an edge

It is natural to see a strong correlation with the line and read it as value. In predictive modelling it is the **opposite**. Measured against the bookie line, the four Infinity variables move with it strongly:

Infinity (Ivan_BP) variable	Correlation with the bookie line *
G_Ivan_BP_NoLineProbability	≈ 0.70
G_Ivan_BP_DefenseProbability	≈ 0.63
G_Ivan_BP_OffenseProbability	≈ 0.59
G_Ivan_BP_NullProbability	≈ 0.53

* Spearman rank-correlation magnitude (0 = unrelated, 1 = identical), vs the bookie moneyline. The four are also 0.45–0.86 correlated with each other — i.e. variations on one Infinity output, not four independent signals.

The “NoLine” paradox. The variable named **NoLineProbability** — the one that, by its name, contains **no** line — is the **most line-correlated of the four (~ 0.70)**. Only two explanations are possible, and we cannot tell which: either Infinity's internal strength model tracks the market so closely that it simply re-derives the line (so it is a redundant copy of a feature we already hold directly, with **no edge**), or the

“NoLine” label does not mean what it says and line information is inside it after all (so it is **circular** — predicting the line from the line). Because we hold only compiled binaries of Infinity, we can never resolve this. The unknowability itself is the risk.

Correlation with the line is not the problem — auditability is. Our own MenOnBase moves with the line about as much (~0.50), and that is fine, because we can prove it is a clean stat reflecting team quality. With a black box we can prove nothing: an identical-looking 0.5–0.7 could be benign team-quality overlap **or** the line leaking in (possibly the *closing* line, known only just before a game), and we have no way to distinguish them. A high, unexplained correlation from an un-inspectable, un-reproducible source is therefore a **red flag, not reassurance** — and a core input we cannot audit or repair degrades silently if Infinity ever changes.

Bottom line: a strong correlation with the line is evidence of duplication or possible circularity, **not** of edge. Genuine edge can only come from signal the line lacks — which is why the trustworthy signals are the ones we compute and can audit ourselves, and why we keep a model that is independent of the line.